

SKIBOTN, Norway

WMO: 010370

Lat: 69.388N

Lon: 20.282E

Elev: 20

StdP: 101.08

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.2	(c) -17.0	(d) -22.7	(e) 0.5	(f) -18.0	(g) -20.5	(h) 0.6	(i) -15.9	(j) 10.1	(k) -5.7	(l) 9.0	(m) -6.2	(n) 2.4	(o) 130	(p) 0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.5	(c) 22.9	(d) 15.0	(e) 20.8	(f) 14.3	(g) 19.0	(h) 13.7	(i) 16.2	(j) 21.0	(k) 15.1	(l) 19.6	(m) 14.1	(n) 18.3	(o) 3.1	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.0	18.4	12.8	9.3	17.3	11.8	8.6	16.1	45.5	21.1	42.4	19.7	39.6	18.4	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 6.8	(c) 6.1	(e) -22.3	(f) 27.5	(g) 2.7	(h) 3.0	(i) -24.3	(j) 29.7	(k) -25.9	(l) 31.4	(m) -27.4	(n) 33.1	(o) -29.4	(p) 35.3		
(a) 7.6	(b) 6.8	(c) 6.1	(e) -22.6	(f) 17.7	(g) 2.7	(h) 1.9	(i) -24.5	(j) 19.1	(k) -26.1	(l) 20.2	(m) -27.6	(n) 21.3	(o) -29.5	(p) 22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-6.3	-6.1	-3.2	1.6	6.5	10.1	13.3	12.0	8.4	2.6	-1.7	-3.6		
	DBStd	8.04	5.65	5.88	4.96	3.59	3.46	2.91	3.13	2.97	3.53	4.68	4.52	5.07		
	HDD10.0	2862	505	452	409	251	118	34	4	12	72	232	350	422		
	HDD18.3	5654	763	685	667	501	366	249	159	196	299	488	600	681		
	CDD10.0	254	0	0	0	0	10	36	107	76	23	3	0	0		
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0		
	CDH23.3	61	0	0	0	0	0	7	44	10	0	0	0	0		
	CDH26.7	12	0	0	0	0	0	1	9	2	0	0	0	0		
(14) Wind	WSAvg	2.7	2.9	3.1	2.8	2.8	2.9	2.7	2.5	2.3	2.5	2.4	2.7	2.8		
(15) Precipitation	PrecAvg	522	58	41	37	21	24	35	56	51	45	57	40	54		
	PrecMax	671	138	100	137	60	64	63	116	106	86	120	91	109		
	PrecMin	370	7	3	4	2	4	4	10	22	10	13	3	14		
	PrecStd	72	36	20	33	13	14	14	24	20	20	28	27	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	8.4	9.9	13.7	21.4	24.1	27.9	24.3	19.7	14.8	11.3	9.5		
		MCWB	4.6	4.6	4.1	7.4	12.4	14.6	18.4	15.8	13.8	9.9	7.8	5.7		
	2%	DB	5.9	6.2	7.4	11.0	18.1	20.1	24.0	21.5	17.3	13.0	8.5	7.2		
		MCWB	3.1	3.3	3.7	6.2	10.9	13.4	16.1	14.8	12.4	9.0	5.6	3.9		
	5%	DB	3.7	4.0	5.5	8.9	15.2	17.6	21.6	19.6	15.8	11.4	6.3	5.2		
		MCWB	1.3	1.8	2.4	4.8	9.2	11.8	15.1	14.3	11.6	8.1	3.8	2.3		
	10%	DB	1.6	2.0	3.7	7.4	12.9	15.8	19.6	17.9	14.2	9.5	4.6	3.3		
		MCWB	-0.3	-0.1	1.0	3.7	7.8	10.8	14.3	13.4	10.5	6.7	2.2	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.9	5.1	4.8	8.2	13.2	15.6	19.1	17.5	14.2	10.4	8.0	6.1		
		MCDB	7.9	7.7	8.5	12.8	19.9	21.5	26.9	21.8	18.4	14.1	11.1	9.1		
	2%	WB	3.2	3.2	3.7	6.4	11.1	13.8	16.9	15.7	12.9	9.2	5.7	4.2		
		MCDB	5.9	6.0	7.1	10.8	17.7	19.1	22.3	19.6	16.7	12.5	8.4	7.0		
	5%	WB	1.5	1.9	2.5	5.0	9.5	12.5	15.7	14.7	11.9	8.2	3.9	2.5		
		MCDB	3.6	3.9	5.3	8.5	14.6	16.9	20.5	18.8	15.2	11.1	6.2	4.9		
	10%	WB	-0.2	0.2	1.3	4.0	8.2	11.3	14.6	13.7	10.9	6.8	2.4	1.0		
		MCDB	1.4	1.9	3.7	7.0	12.5	14.8	18.9	17.5	13.7	9.2	4.4	3.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	6.9	8.4	8.2	8.7	7.8	8.5	8.9	8.1	6.5	6.1	6.3		
		MCDBR	7.7	7.4	7.1	9.7	13.6	12.1	13.1	12.4	9.8	7.2	7.2	6.7		
		MCWBR	6.1	5.7	4.7	6.1	7.4	6.4	6.3	6.7	5.9	4.8	5.2	5.0		
		MCWBR	7.6	7.8	7.0	9.4	13.1	10.8	11.7	11.4	9.0	7.2	7.2	6.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.216	0.265	0.316	0.337	0.350	0.352	0.357	0.343	0.314	0.272	N/A	N/A		
		taud	1.901	2.021	2.052	2.118	2.216	2.396	2.441	2.465	2.497	2.166	N/A	N/A		
		Ebn at Noon	19	451	666	786	827	842	825	794	719	448	N/A	0		
		Edh at Noon	4	45	86	110	113	98	91	79	58	38	N/A	0		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.01	0.33	1.59	3.53	4.88	4.89	4.50	3.19	1.68	0.51	0.06	0.00		
		RadStd	0.00	0.05	0.14	0.24	0.36	0.46	0.40	0.25	0.15	0.06	0.01	0.00		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page